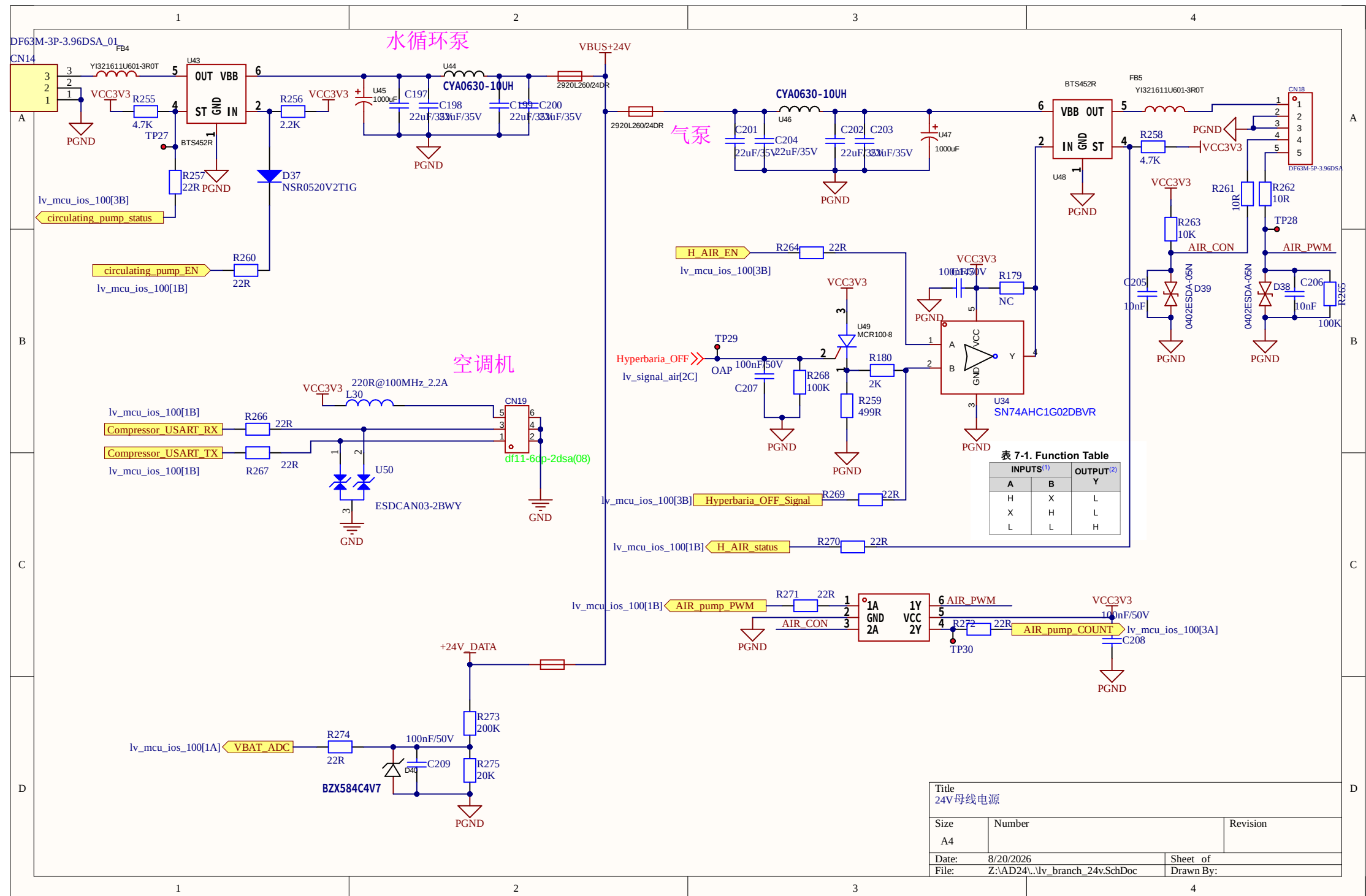
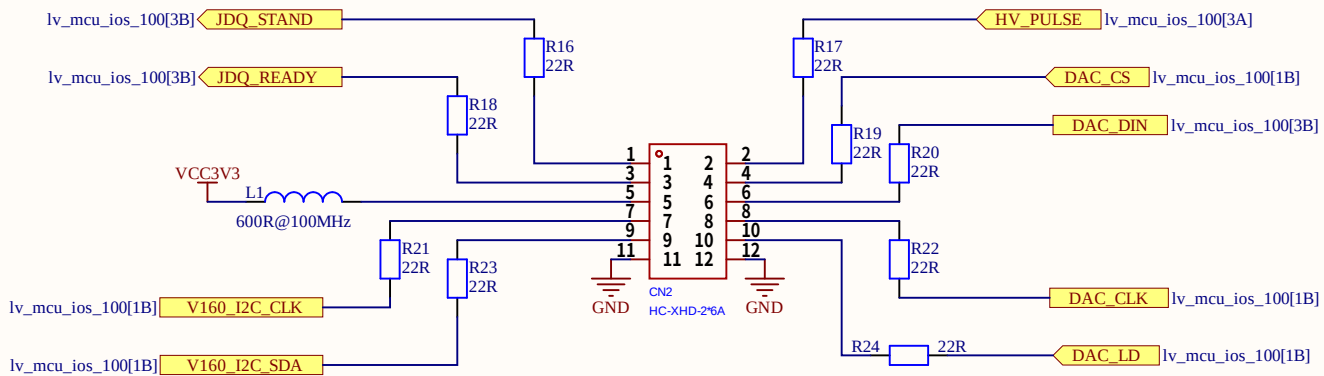
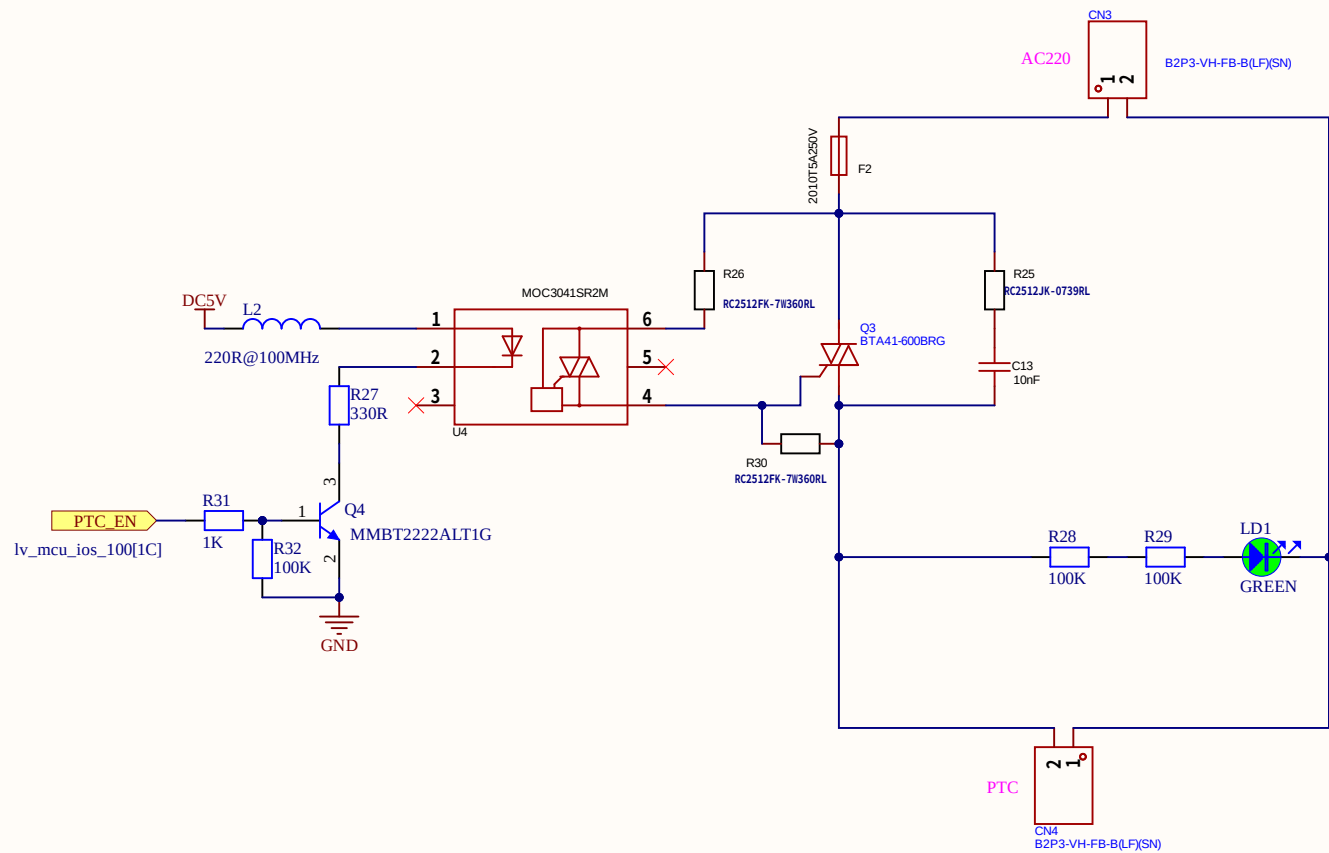




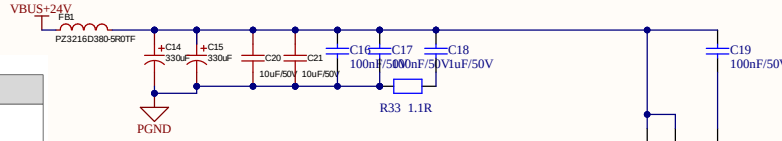


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File:	Z:\AD24\lv_power_diode_laser_cc.SchDoc	Drawn By:

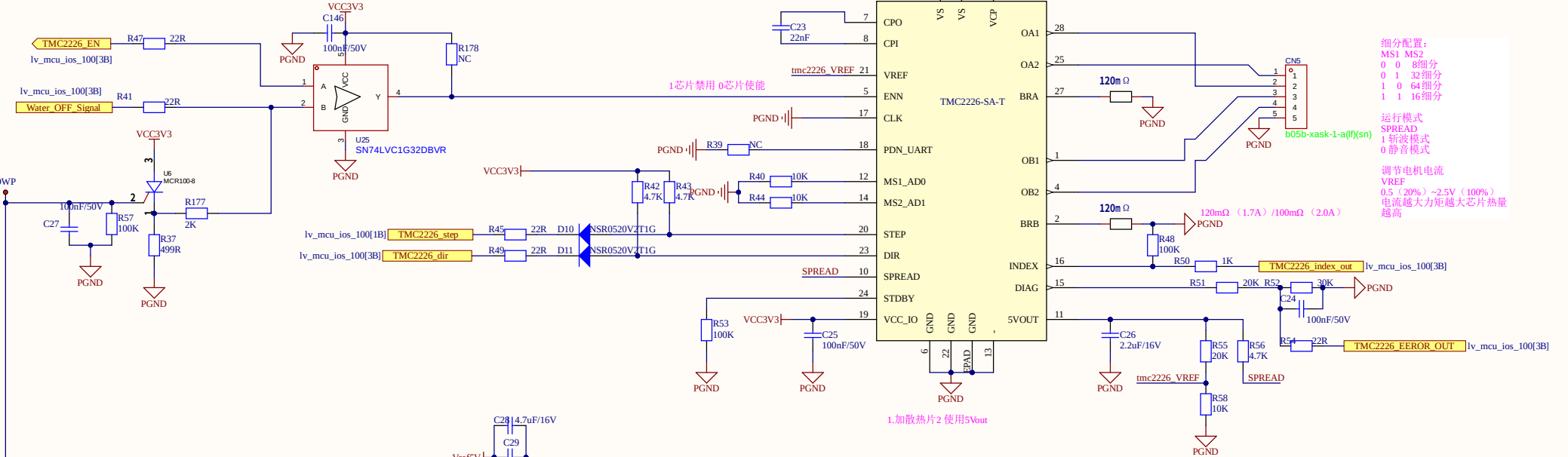


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Function Table		
INPUTS		OUTPUT Y
A	B	Y
H	X	H
X	H	H
L	L	L



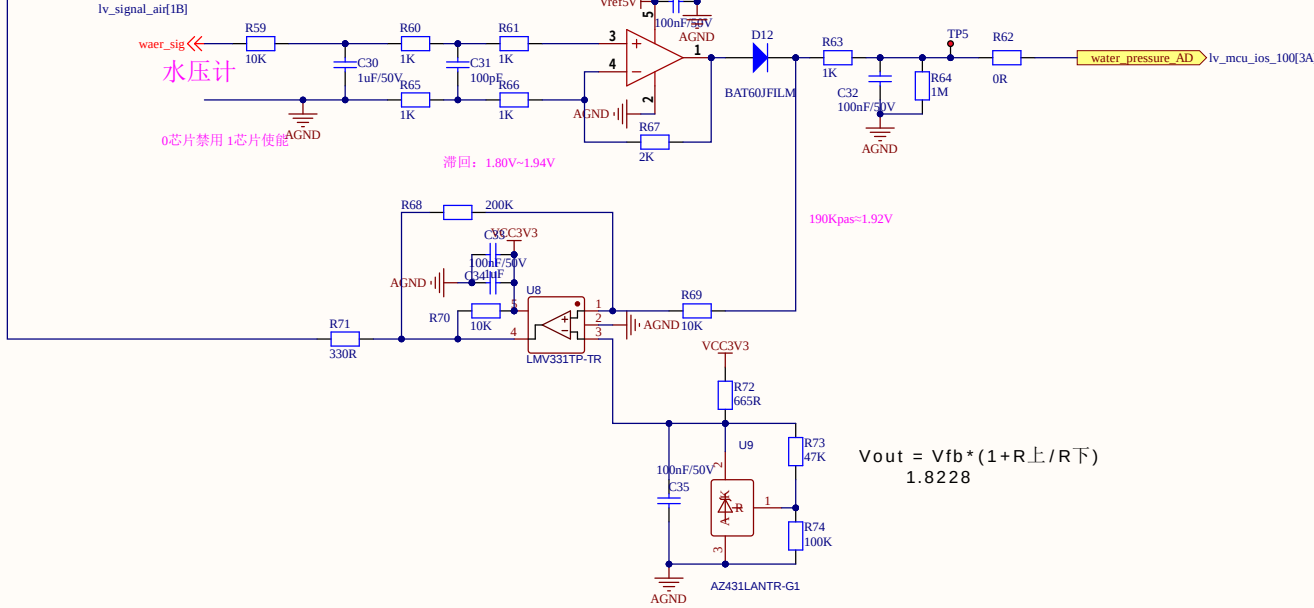
蠕动泵电机



细分配置:
MS1 MS2
0 0 8细分
0 1 32细分
1 0 64细分
1 1 16细分

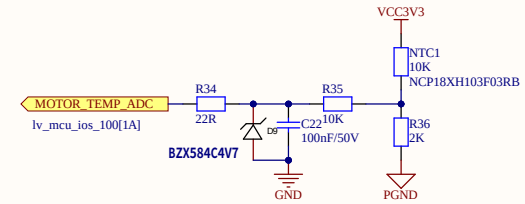
运行模式
1 斩波模式
0 静音模式

调节电机电流
VREF
0.5 (20%) ~2.5V (100%)
电流越大扭矩越大芯片热量越高

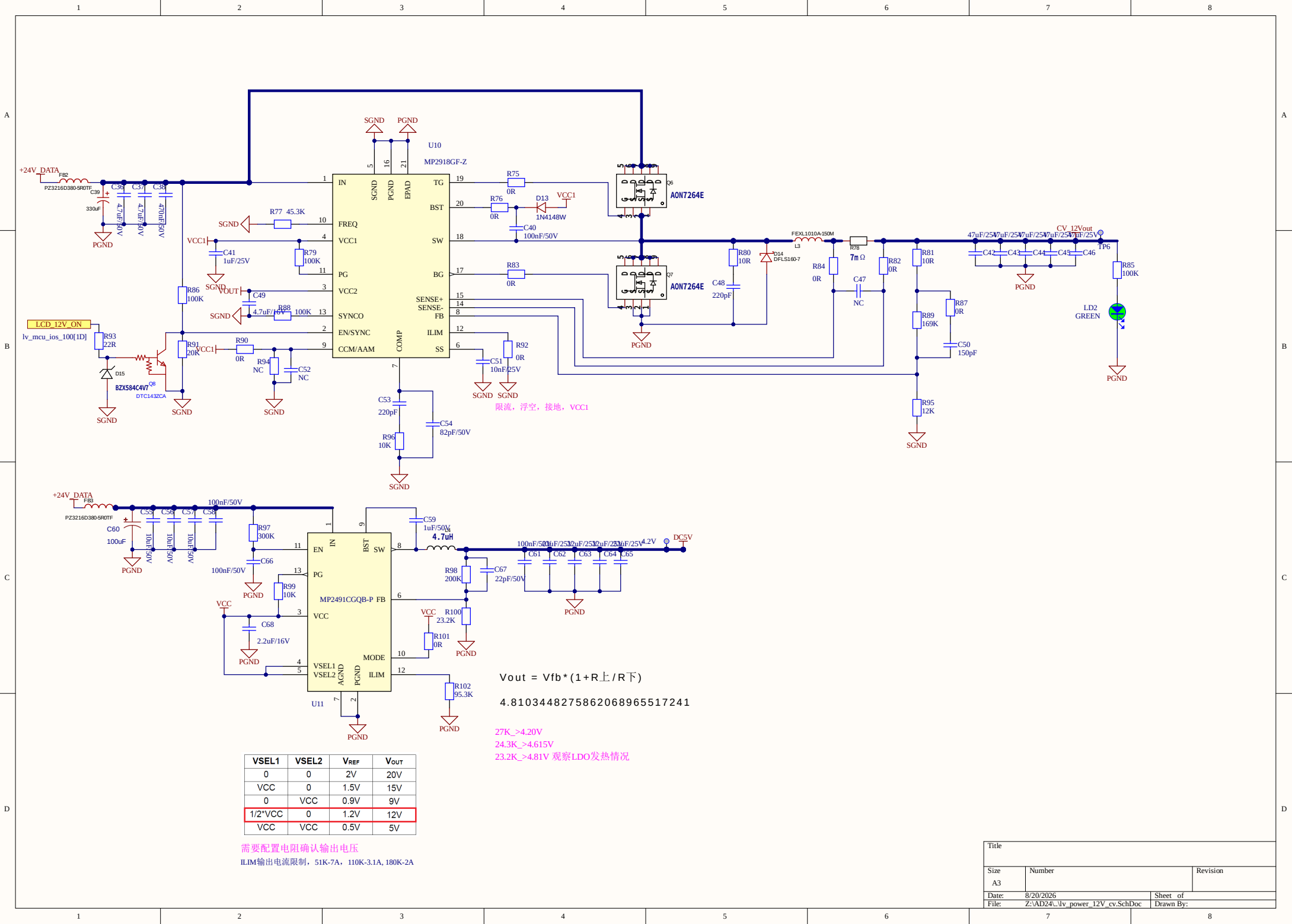


$$V_{out} = V_{fb} * (1 + R_{上} / R_{下})$$

1.8228



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VSEL1	VSEL2	VREF	VOUT
0	0	2V	20V
VCC	0	1.5V	15V
0	VCC	0.9V	9V
1/2*VCC	0	1.2V	12V
VCC	VCC	0.5V	5V

需要配置电阻确认输出电压
ILIM输出电流限制，51K-7A，110K-3.1A，180K-2A

$V_{out} = V_{fb} * (1 + R_{上} / R_{下})$
4.8103448275862068965517241
27K > 4.20V
24.3K > 4.615V
23.2K > 4.81V 观察LDO发热情况

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1

2

3

4

A

A

B

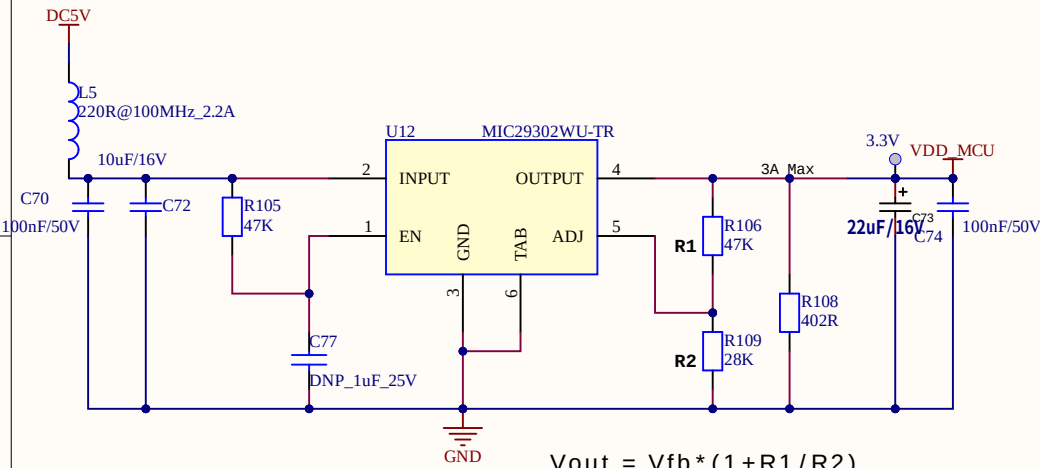
B

C

C

D

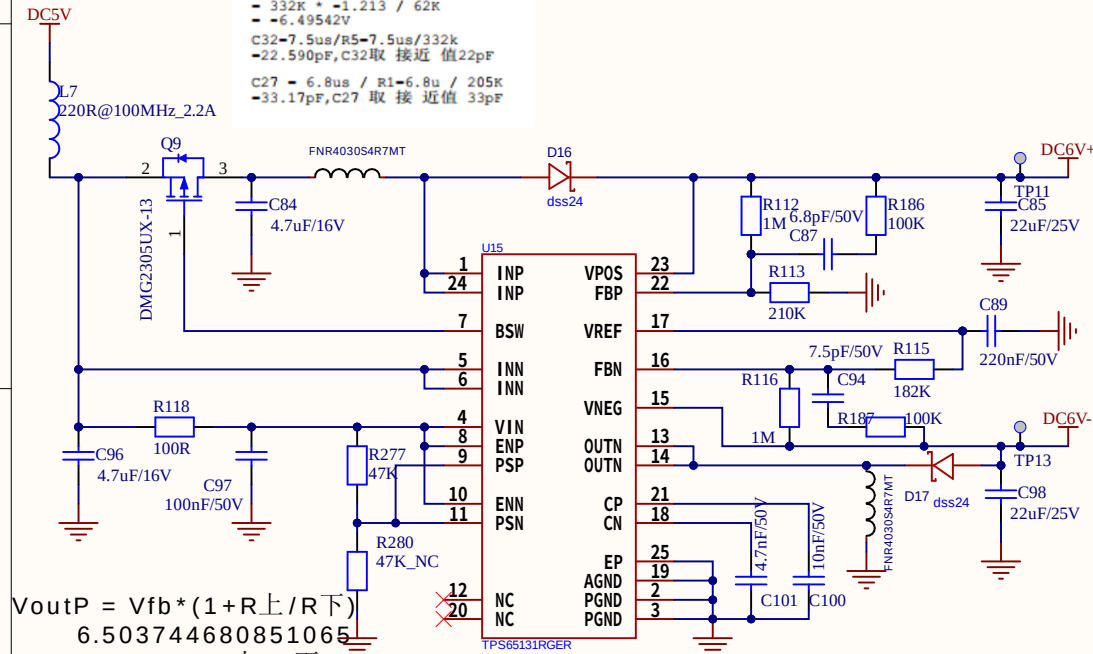
D



$$V_{out} = V_{fb} * (1 + R1/R2)$$

$$3.3214285714285716$$

AVDD_External
 - (R1 / R2 + 1) * 1.213
 - (205K / 47K + 1) * 1.213
 - 6.503744V
 AVEE_External
 - R5 * -1.213 / R4
 - 332K * -1.213 / 62K
 - -6.49542V
 C32 = 7.5us / R5 = 7.5us / 332k
 = 22.590pF, C32取 接近 值 22pF
 C27 = 6.8us / R1 = 6.8us / 205K
 = 33.17pF, C27取 接近 值 33pF

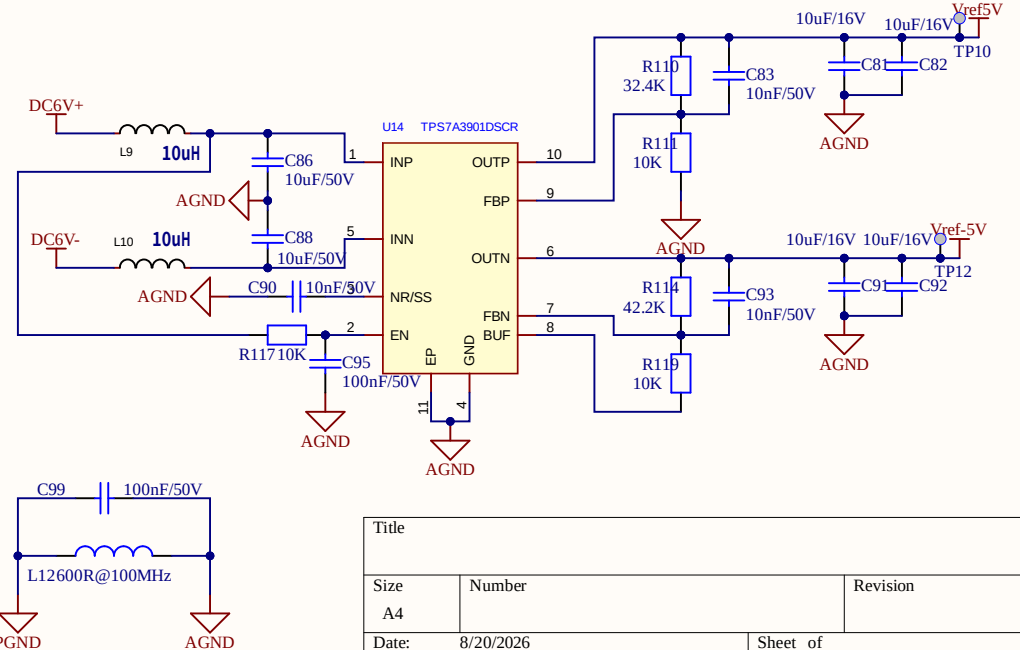
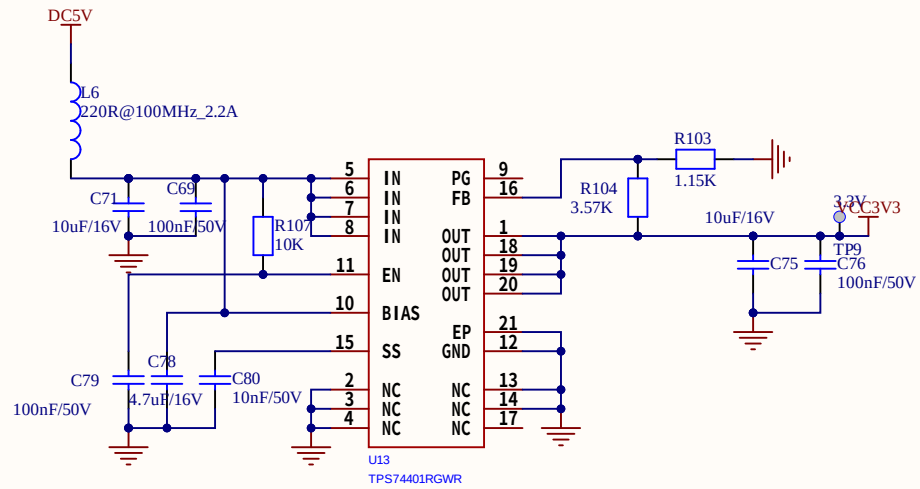


$$V_{outP} = V_{fb} * (1 + R_{上} / R_{下})$$

$$6.503744680851066$$

$$V_{outN} = V_{fb} * (R_{上} / R_{下})$$

$$-6.49541935483871$$



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1. $R_{sense} = V_{ref} / I_{led}$
= $0.1V / 0.05mA$
= 2
2. 上电顺序: 上电 -> pwm
3. PWM控制信号:
<1V 关机
1V~2.07V PWM调光模式
>2.07V 模拟调光模式
4. PWM频率>10KHZ, 低频率纹波大

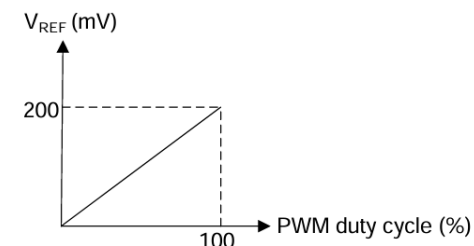
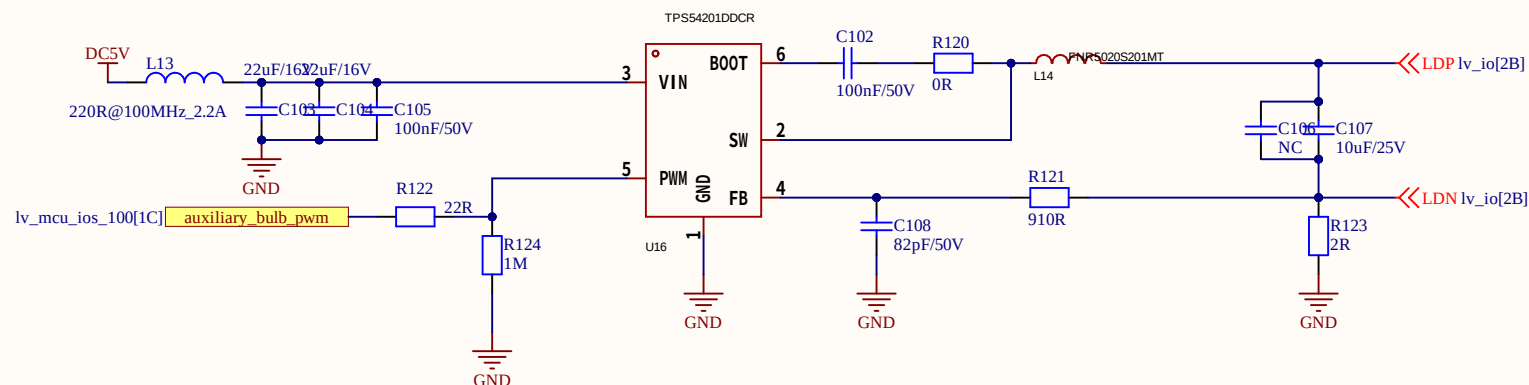
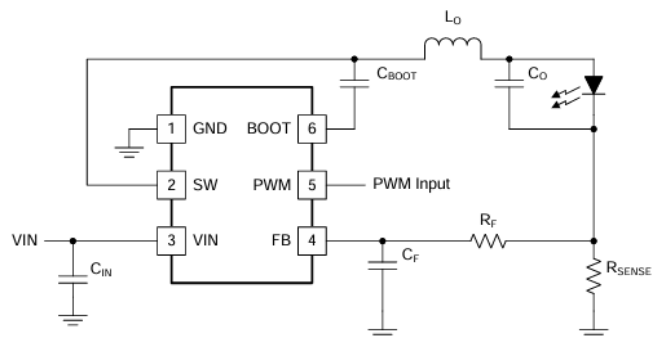


图 16. V_{REF} vs PWM Duty Cycle in Analog Dimming Mode

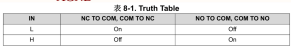


简化电路原理图



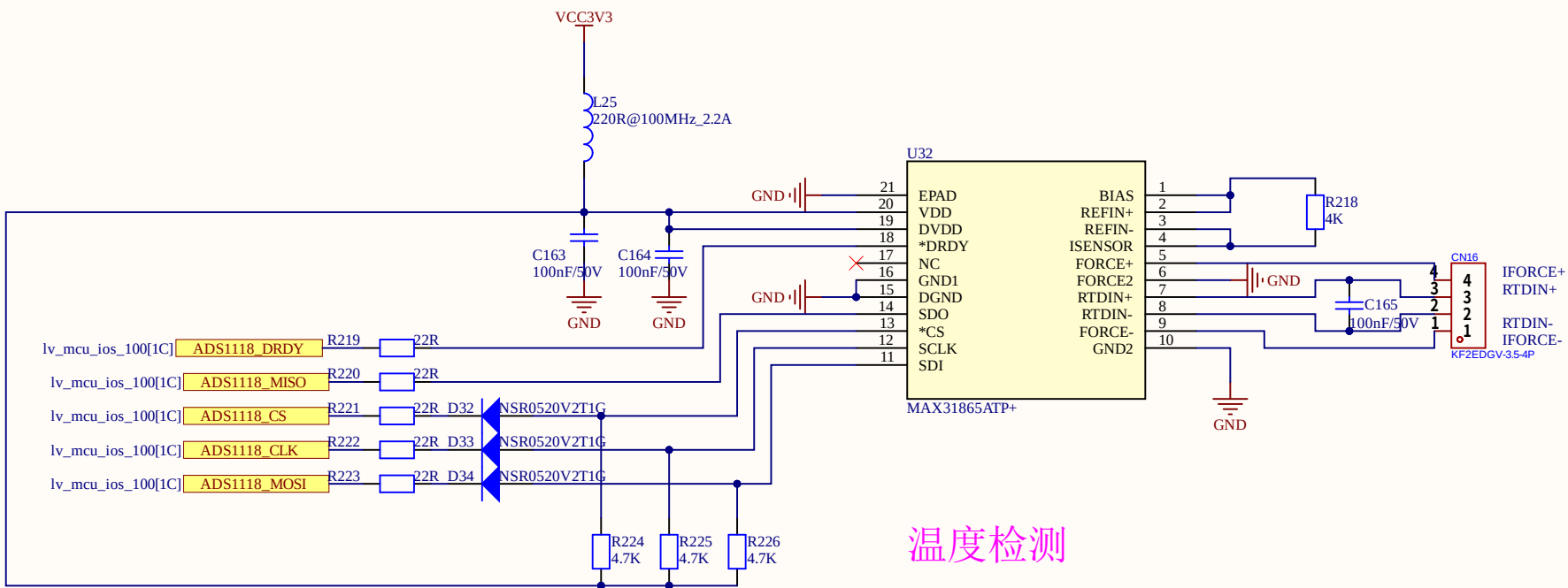
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Title 指示灯电源		
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File: Z:\AD24\lv_power_indicator_light_cc\	Drawn By: SDD	



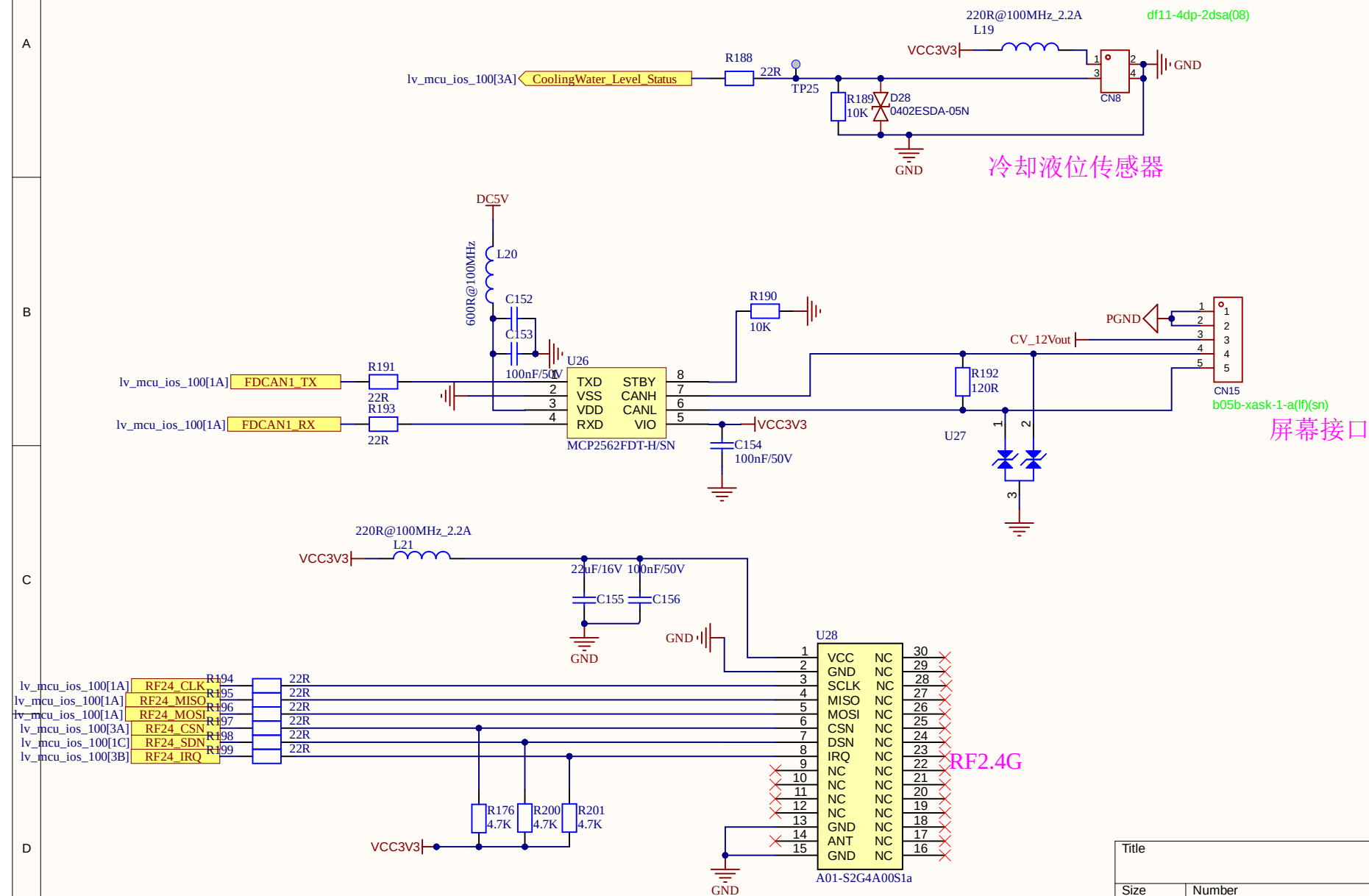
IN	NC TO COM, COM TO NC	NO TO COM, COM TO NO
L	On	Off
H	Off	On

Title		
Size A3	Number	Revision
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File: Z:\AD24\...lv_signal_link.SchDoc	Drawn By:	

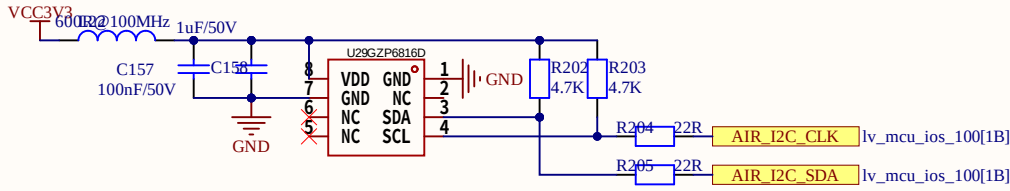


温度检测

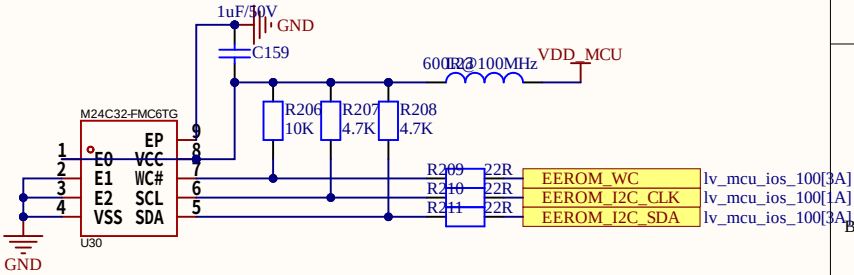
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Date:	8/20/2026	Sheet of
File:	Z:\AD24\lv_signal_temp.SchDoc	Drawn By:



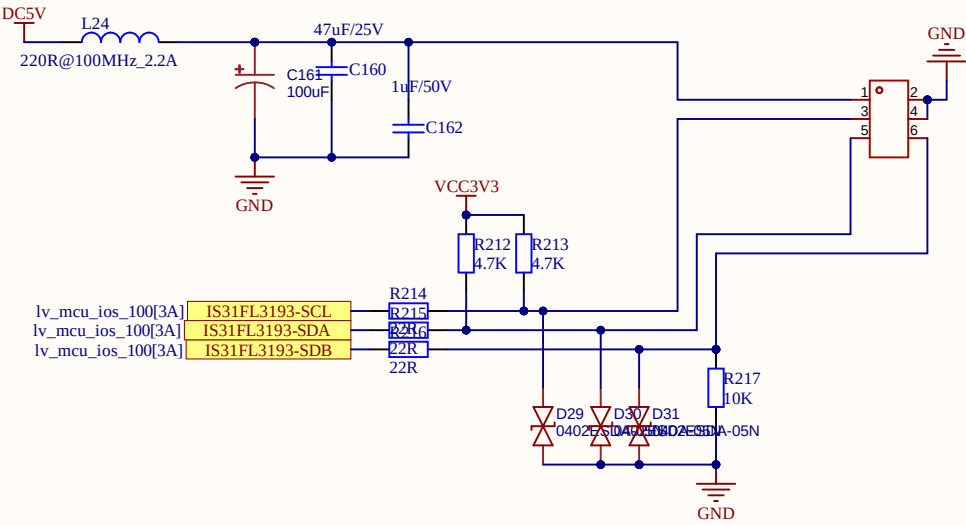
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环境气压、温度传感器



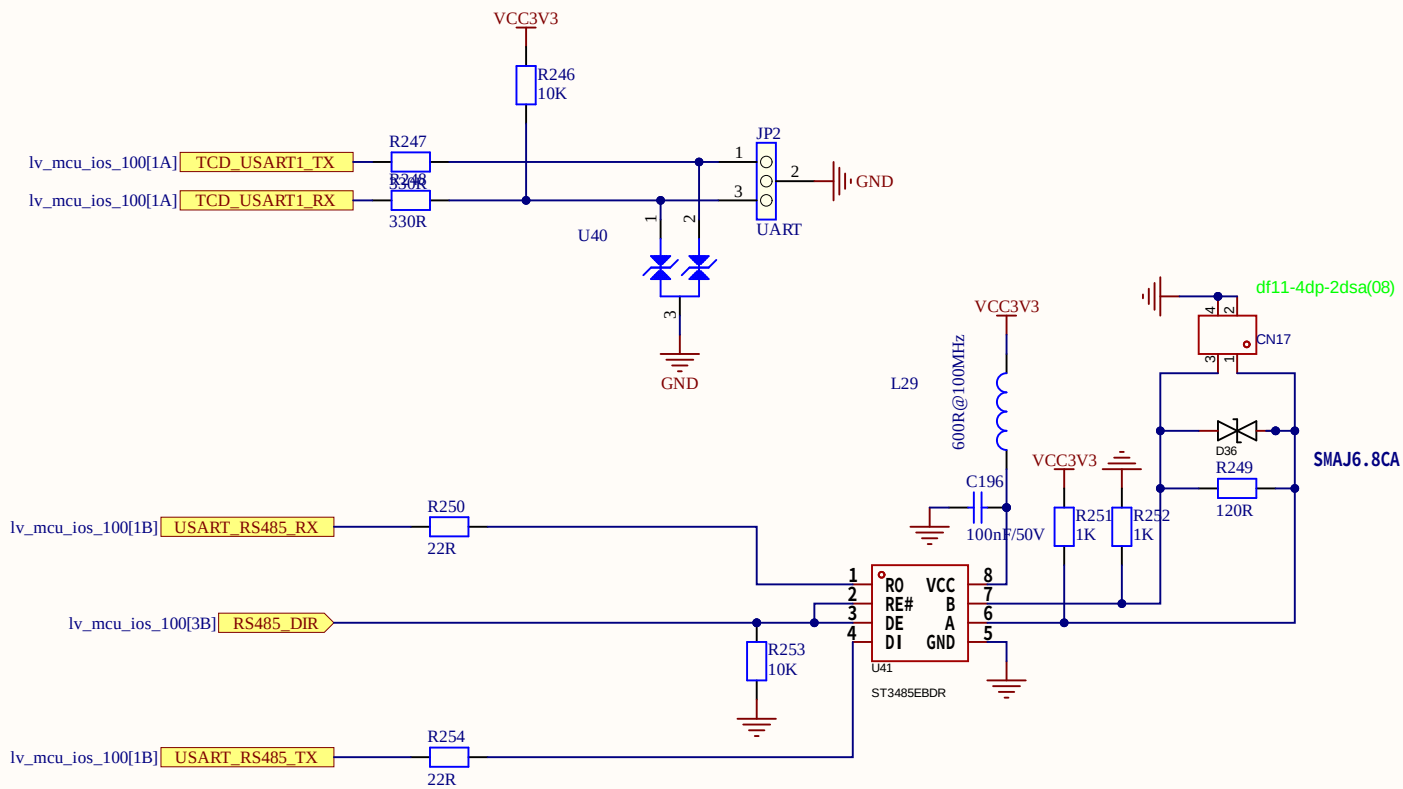
EEROM



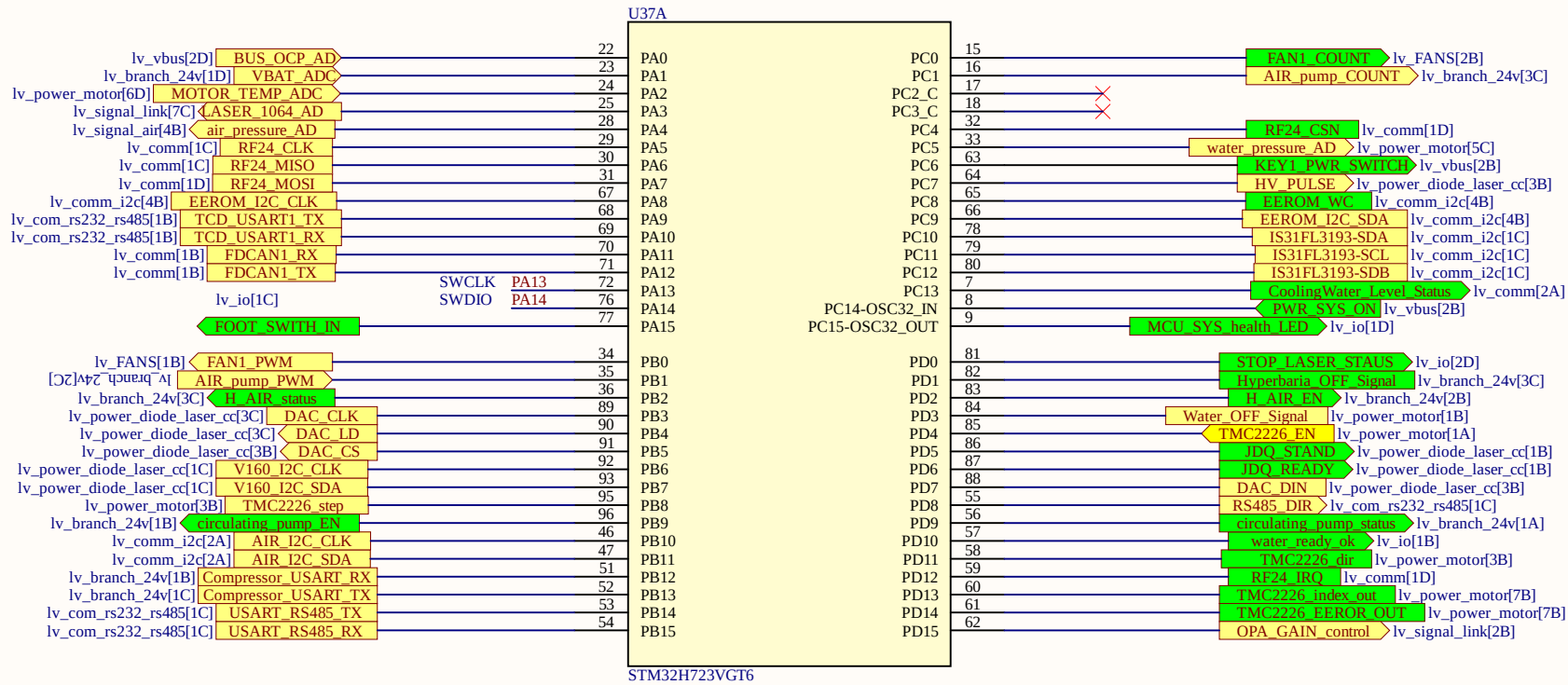
RGB氛围灯

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Size	Number	Revision
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Date:	8/20/2026	Sheet of
File:	Z:\AD24\lv_comm_i2c.SchDoc	Drawn By:

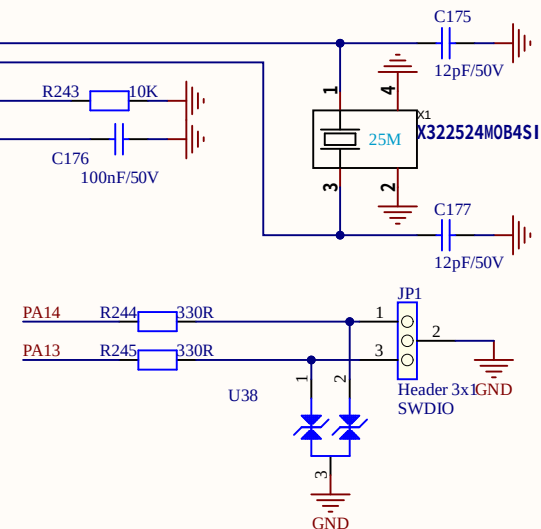
RS232 & RS485 (USART1)



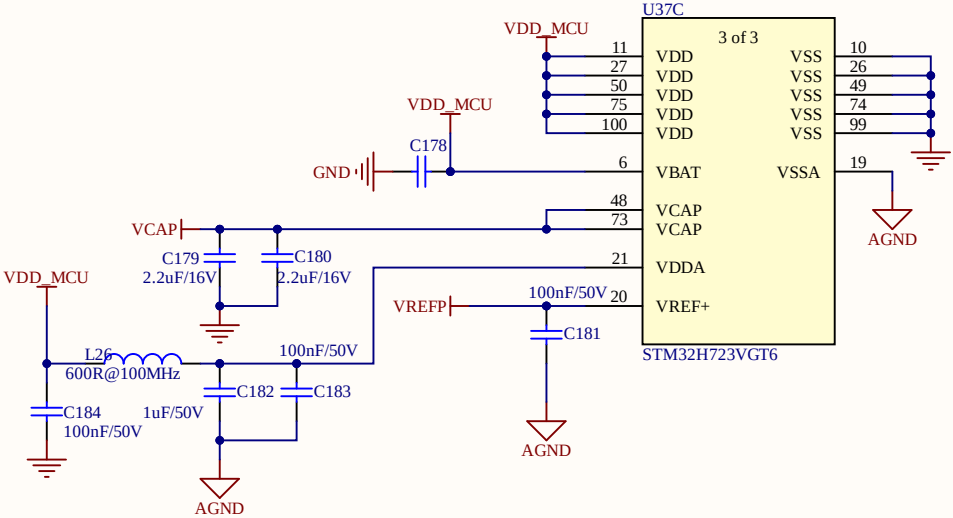
Title		
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File:	Z:\AD24\..lv_com_rs232_rs485.SchDoc	Drawn By:



- 1.删除风扇PC1,PB2,删除电磁阀驱动和状态引脚
 - 2.增加调速气泵PB2 PWM调速
 - 3.PB9改到PE0, PE13改到PB9
 - 4.pc15改到PE13
 - 5.PA15改到PE14
 - 6.把PE7改到PA15
 - 7.心跳LED改为PC15
 - 8.遥控连锁改为PE15
- 20260409

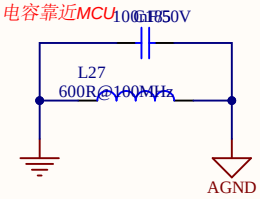
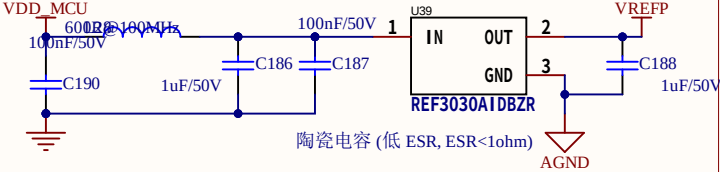
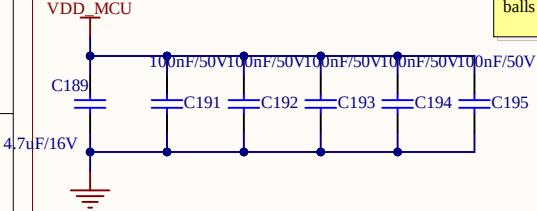


MCU PWR SUPPLIES

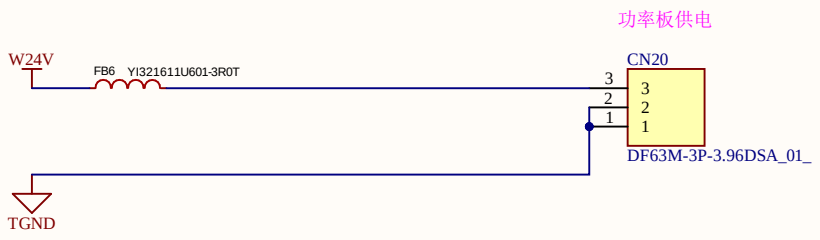
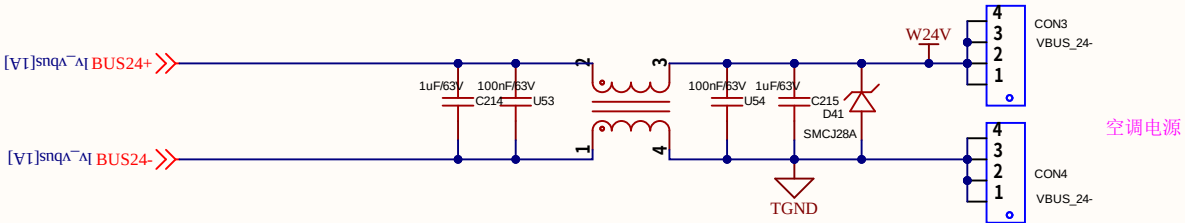


MCU DECAPS
Ceramic capacitor (Low ESR, ESR<1ohm)

capa near balls

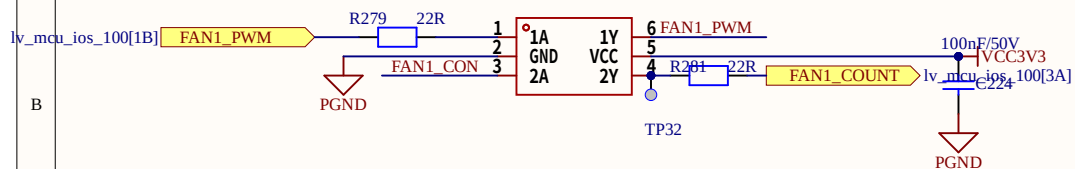
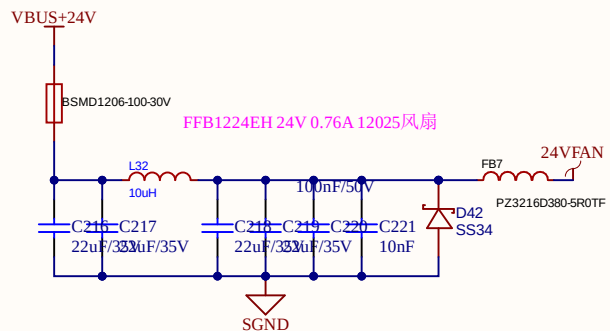


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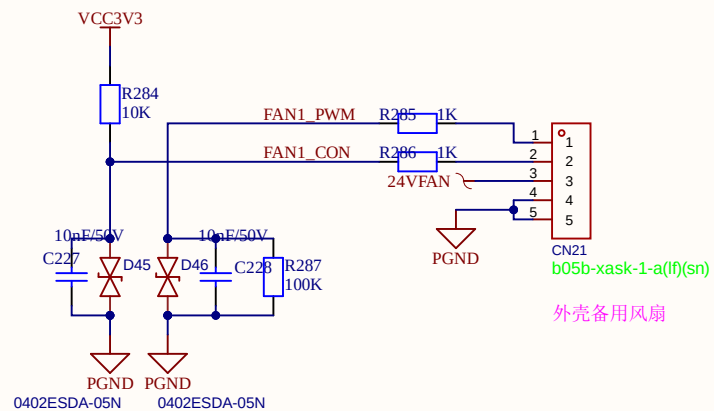
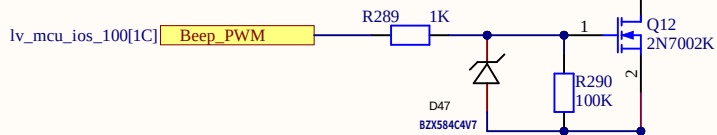
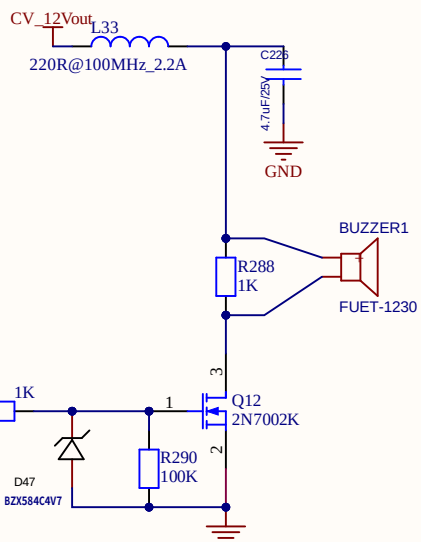


功率板上控制IO和3.3V隔离

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BEEP



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